QUANTUM COMPUTING HACKATHON PAKISTAN 2026

February 6-8, 2026

Beyond Classical Limits: Quantum Route Revolution

Quantum Enhanced Solution for Logistic Route Optimization.

TEAM D: ENTANGLED MINDS

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Problem: Pakistan's Logistics Nightmare

THE CRISIS: Annual Impact

Urban Network

Rs 2.3 Billion Annual Loss

Logistics Crisis

40%
Fuel Waste

CONSTRAINED DELIVERY ROUTE (TSP Variant)

Delivery Vehicle

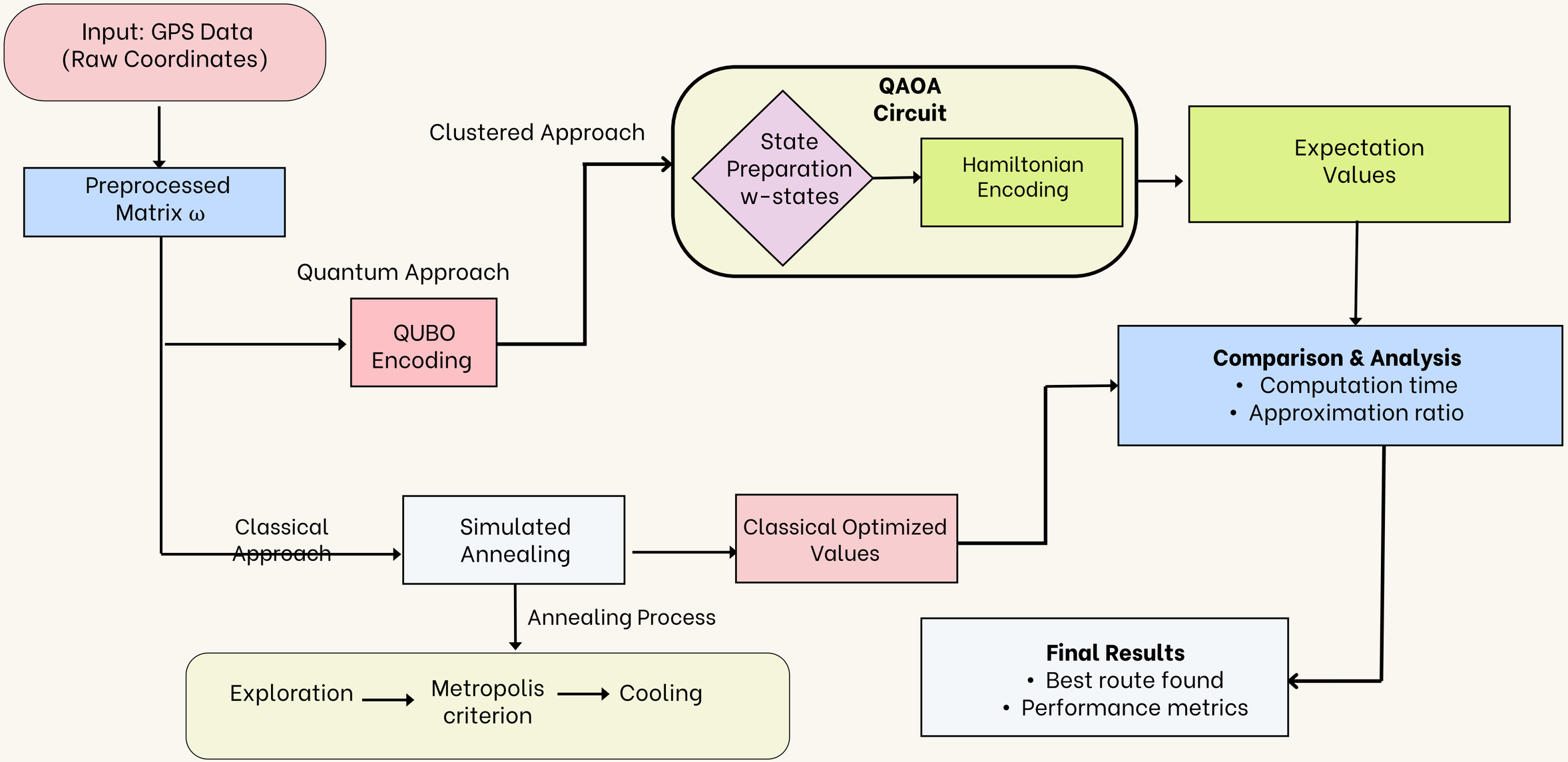
~150
Daily Stops

Optimization

Network

WORKFLOW: HYBRID CLASSICAL-QUANTUM APPROACH

Team D

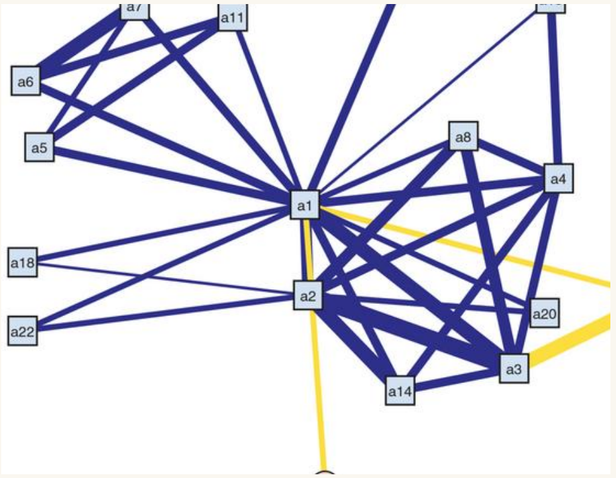


PHASE 1

Raw Data → Structured Data

GPS Points

→



Road Network Graph

→

$$\begin{array}{c|cccc} & A & B & C & D \\ \hline A & 0 & X & X & X \\ B & X & 0 & X & X \\ C & X & X & 0 & X \\ D & X & X & X & 0 \end{array}$$

Distance Matrix

Input: Raw Data

- GPS coordinates of delivery points
- Road speeds, one-way streets, intersections

Output: Matrix ω

- n×n matrix: **ω {i,j}** = distance from city i to city j.

QUBO Encoding C(x)

Input: Structured Data

$$w \in R^{(n \times n)}$$

$$w_{i,j} =$$ distance from node i to j

QUBO Encoding

$$D(x) = \sum_{i,j=1}^{n} w_{i,j} \sum_{t=1}^{n-1} x_{i,t} x_{j,t+1}$$

Add Real-World Constraints

$$C(x) = D(x) + P(x) + T(x)$$

- $D(x)$: Distance objective
- $P(x)$: Each node once
- $T(x)$: Time windows

Logistics Constraints

Constraint 1: Each Node Visited Exactly Once

$$P(x) = \lambda_p \underbrace{\sum_i \left(\sum_t x_{i,t} - 1 \right)^2}_{\text{each city visited once}} + \lambda_q \underbrace{\sum_t \left(\sum_i x_{i,t} - 1 \right)^2}_{\text{one city per time step}}$$

Constraint 2: Time step constraints

$$T_{i,t} = \begin{cases} 1 & \text{if city } i \text{ is not allowed to be visited at step } t, \\ 0 & \text{otherwise.} \end{cases}$$

$$T(x) = \lambda_T \sum_{i,t} T_{i,t} x_{i,t},$$

Logistics Constraints

Constraint 3: Binary Node Compatibility (BNC)

$$K(x) = \lambda_k \sum_{\substack{i,j,t \\ i \neq j}} x_{i,t} x_{j,t+1} (1 - k_i \oplus k_j),$$

$$k_i = \begin{cases} 1 & \text{if node } i \text{ belongs to category A,} \\ 0 & \text{if node } i \text{ belongs to category B.} \end{cases}$$

$$\tilde{\omega}_{i,j} = \omega_{i,j} + \lambda_k (1 - k_i \oplus k_j).$$

Constraint 4: Road Related Constraints

$$R_{i,j} = \begin{cases} 1 & \text{if node } i \text{ cannot be directly connected to node } j, \\ 0 & \text{otherwise.} \end{cases}$$

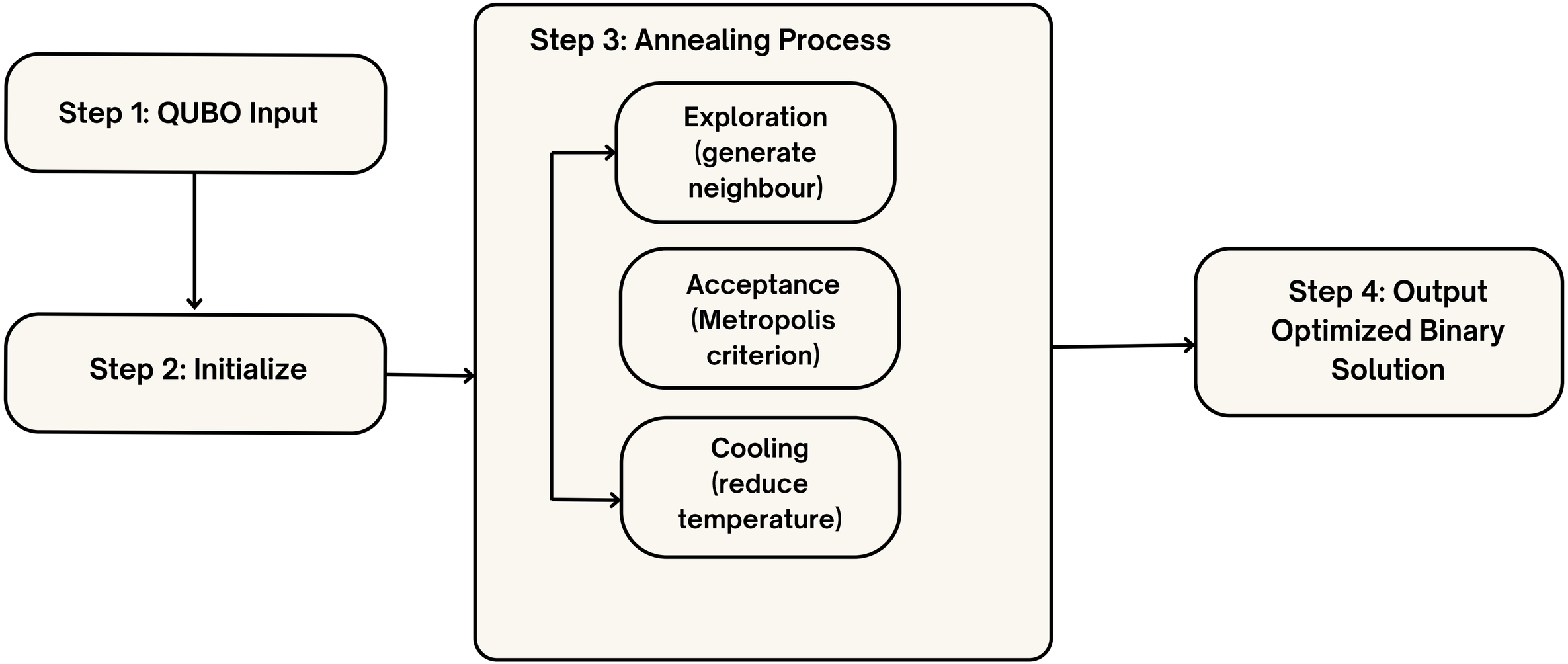
$$R(x) = \lambda_R \sum_{\substack{i,j,t \\ i \neq j}} x_{i,t}\, x_{j,t+1}\, R_{i,j}.$$

$$\tilde{\omega}_{i,j} = \omega_{i,j} + \lambda_R R_{i,j}.$$

PHASE 3

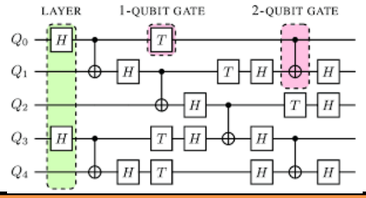
Simulated Annealing Process

Team D



QAOA (Quantum Approximate Optimization Algorithm)

PHASE 4



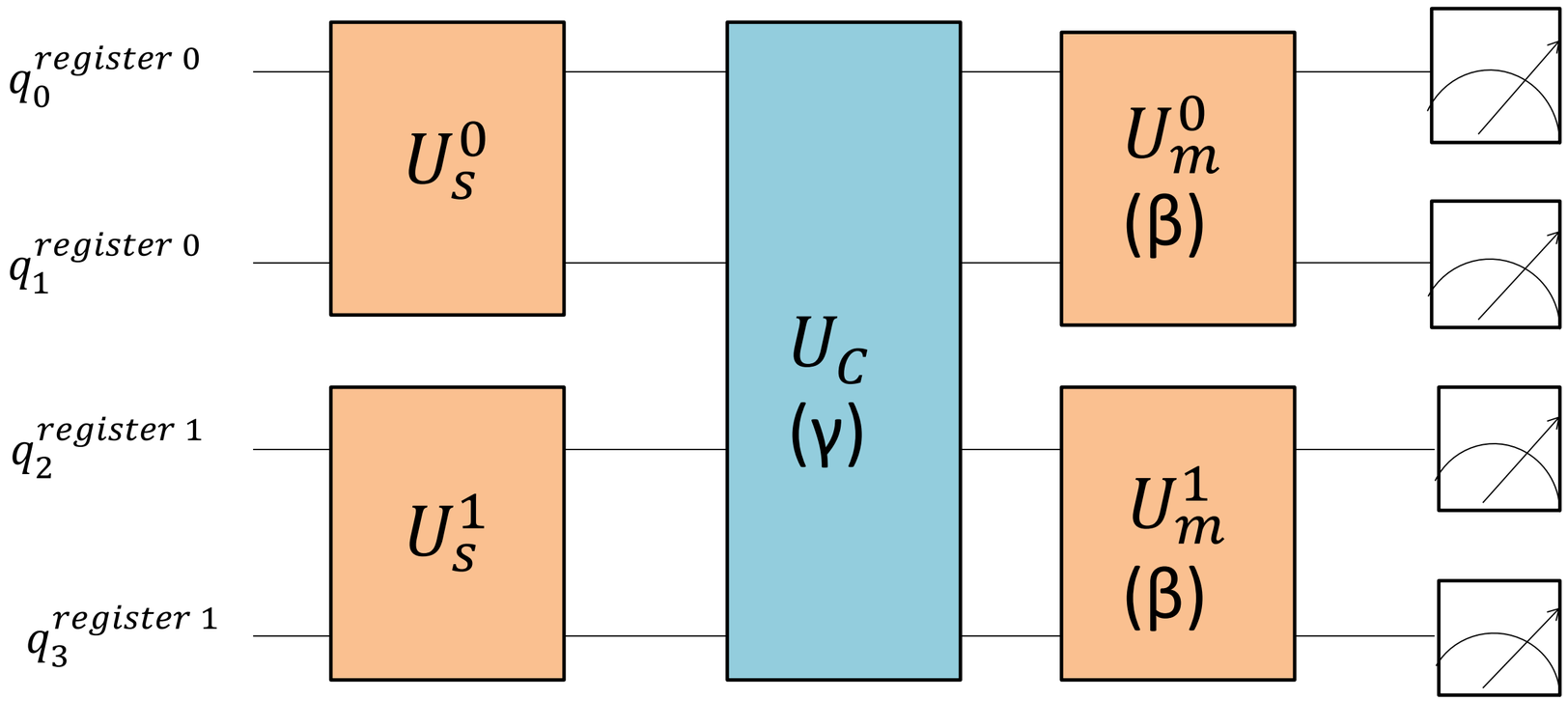
Quantum Circuit

γ, β

$$\langle H_C \rangle_{\gamma,\beta} = \langle \vec{\gamma}, \vec{\beta} | H_c | \vec{\gamma}, \vec{\beta} \rangle$$

Classical Optimizer

Quantum Circuit

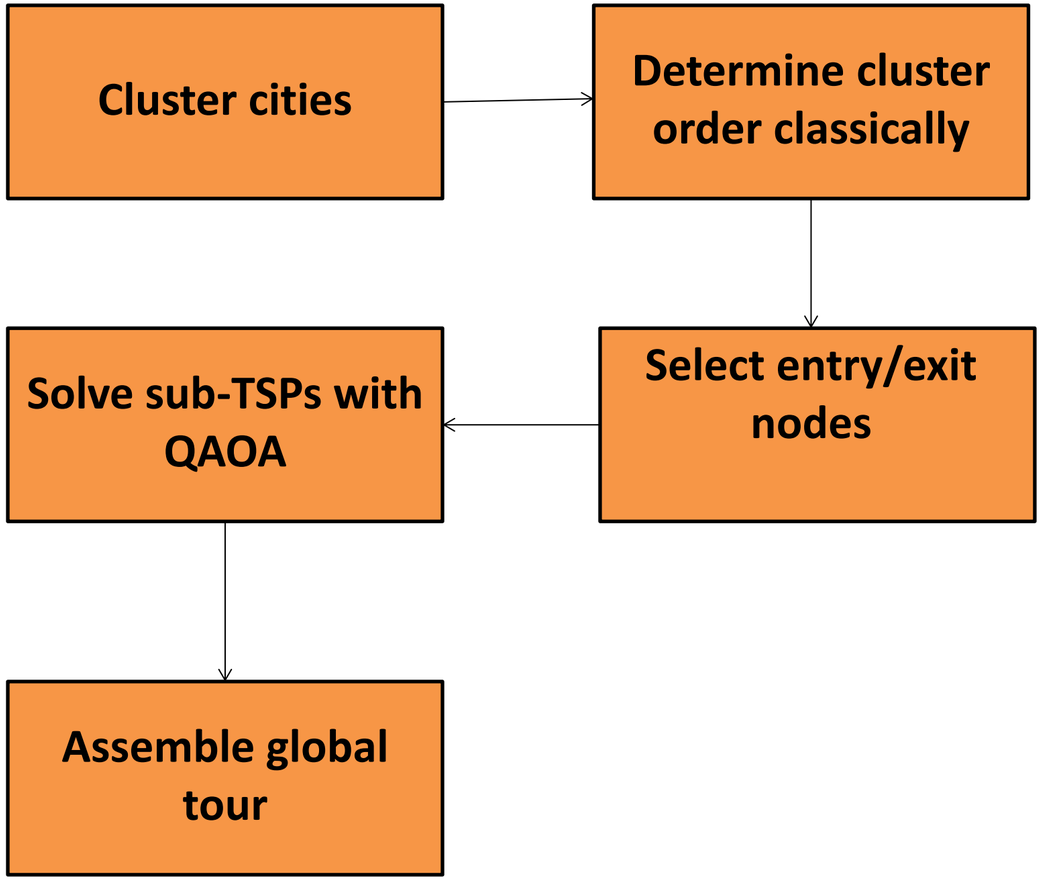


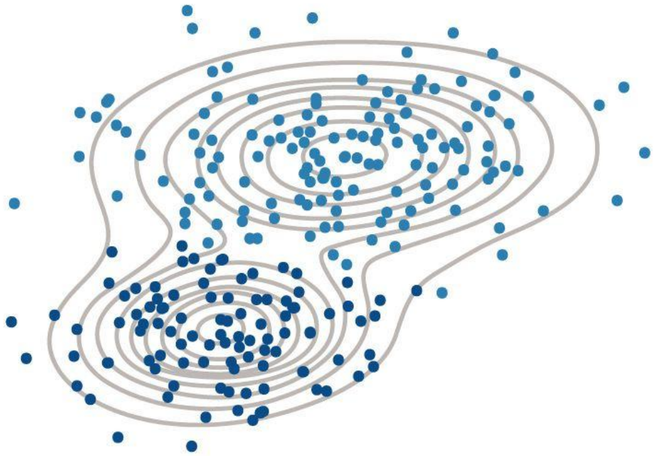
$$U_s^t|0\rangle^{\otimes 3} = |D_1^3\rangle = \frac{1}{\sqrt{3}}\left(|100\rangle + |010\rangle + |001\rangle\right)$$

$$U_c(\gamma) = e^{-i\gamma H_c}$$

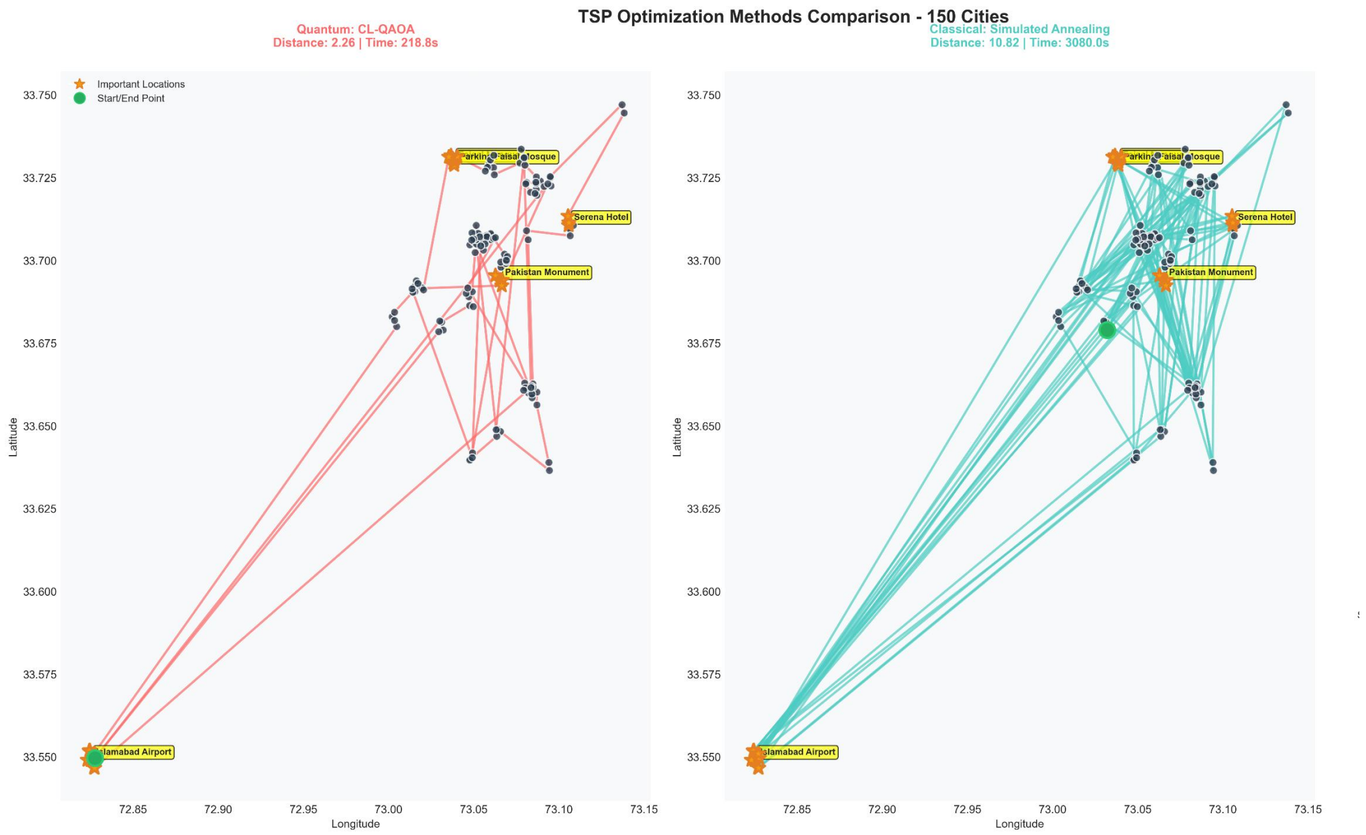
$$U_m^t(\beta_i) = U_s^t\left(I_d - \left(1 - e^{-i\beta_i}\right)|0\rangle\langle 0|\right)(U_s^t)^\dagger$$

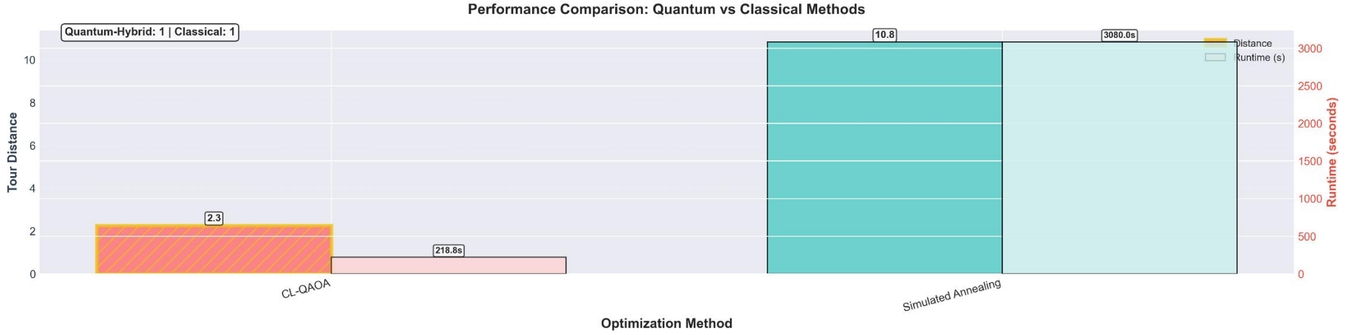
Clustering QAOA (Cl-QAOA)





Results


TSP Optimization Methods Comparison - 150 Cities
Quantum: CL-QAOA
Distance: 2.26 | Time: 218.8s
Classical: Simulated Annealing
Distance: 10.82 | Time: 3080.0s
Important Locations
Start/End Point
Serena Hotel
Pakistan Monument
Latitude
Longitude


Performance Comparison: Quantum vs Classical Methods
Quantum-Hybrid: 1 | Classical: 1
Tour Distance
Runtime (seconds)
Distance
Runtime (s)
2.3
218.8s
10.8
3080.0s
CL-QAOA
Simulated Annealing
Optimization Method

Comparison & Analysis

```
📊 PERFORMANCE COMPARISON:
+---------------------+------------------+------------+-----------+--------------+---------------------------+
|       Method        |       Type       |  Distance  |  Runtime  |  Norm. Cost  |         Tour Start        |
+=====================+==================+============+===========+==============+===========================+
|       CL-QAOA       | ⚛️ QUANTUM-HYBRID |       2.26 |  218.80s  |    1.000x    | [119, 100, 19, 110, 36]... |
+---------------------+------------------+------------+-----------+--------------+---------------------------+
| Simulated Annealing |   🖥️ CLASSICAL    |      10.82 | 3080.05s  |    4.794x    |  [19, 38, 14, 26, 59]...   |
+---------------------+------------------+------------+-----------+--------------+---------------------------+

⭐ BEST PERFORMANCE: CL-QAOA - 2.26
📉 WORST PERFORMANCE: Simulated Annealing - 10.82

⚛️ vs 🖥️ QUANTUM-CLASSICAL COMPARISON:
   Best Quantum-Hybrid: 2.26
   Best Classical: 10.82
   Relative Performance: 0.209x
   ✅ CL-QAOA outperforms classical by 79.1%

💾 Comprehensive results saved to: hackathon_summary_150cities_20260205_220957.json
📈 CSV data saved to: results_comparison_150cities_20260205_220957.csv
```

Future Perspectives

1. Extend Framework & Expand Applications
2. Optimize Quantum Resource Efficiency
3. Enhance the Clustering Phase
4. Experimental Validation on Quantum Hardware
5. Executing Quantum Optimization at Scale

Thank You.